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The yield stress of concentrated flocculated suspensions of size distributed particles

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Synopsis

An investigation of shear yield stress is made on well-characterized alumina suspensions of different distributed particle sizes at the vicinity of the particle isoelectric point (IEP) across a wide range of volume fractions. Experimental results are compared with recently developed models [Kapur *et al.* (1997); Scales *et al.* (1997)] and structural effects on the yield stress are examined. The models predict the magnitude order of the yield stress below a volume fraction of approximately 0.42, suggesting that interparticle forces play a dominant role in determining the network strength in this concentration region. Deviations between experimental results and theoretical predictions are explained in terms of structural effects being controlled by a competition between weak particle–particle linkages and geometric resistance on the network strength. At higher volume fraction, the effect of geometric resistance on the deformation of suspensions becomes more pronounced. A number of models for the yield stress of size distributed suspensions are then proposed. Results suggest that the effect of polydispersity of particles on the yield stress of suspensions can be well characterized by a surface area average diameter and the broad size distributed suspension exhibits a higher yield stress than the narrow size distributed suspension of the same volume average diameter. © *1999 The Society of Rheology.* [S0148-6055(99)00303-X]

I. INTRODUCTION

The efficient utilization, transport, and handling of concentrated particulate suspensions requires the ability to predict the rheological properties as a function of the various physical parameters, including solids loading, particle size and size distribution, and surface chemistry properties of the particles. In many situations pertaining to the processing of flocculated concentrated suspensions, yield stress turns out to be of greater significance than viscosity.

Although a large amount of theoretical and experimental work has been carried out on yield stress, earlier models relating yield stress and modulus with particle volume concentration and size are somewhat unsatisfactory, especially for the systems of distributed particle sizes. While yield stress at a fixed concentration appears to vary inversely with the square of particle size or of some percentile size such as 10% and 50% [Thomas (1961); Buscall *et al.* (1987); Leong *et al.* (1995)], the network models predict an inverse dependence on particle size [Tempel (1961) and (1979)] and the cell model predicts an inverse cube dependence at the isoelectric point (IEP) [Tanaka and White (1980)]. On the other hand, the floc and elastic floc models predict a floc size dependence [Micheals and

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Bolger (1962); Hunter and Nicol (1968); Firth (1976); Firth and Hunter (1976)], and Doi and Chen (1989) suggest that the cluster size is not the key parameter for determining the rheological properties of flocculated suspensions.

The yield stress increases very sharply with an increase in the solid loading. The simple power-law equation for correlating yield stress with particulate volume fraction showed an exponent of 3 [Thomas (1961)], or of 4 [Leong *et al.* (1995)] or 2 reported by Wildemuth and Williams (1985). It is important to note that the surface chemistry conditions of the aqueous suspensions used in most of the above studies were not clearly defined, moreover, the particle size distribution which surely must play a significant role in determining the rheological properties of suspensions was not included in the above models.

Kapur *et al.* (1997) recently developed a model for the yield stress of concentrated, flocculated, size distributed particulate suspensions. The model estimates the yield stress utilizing input values of the system Hamaker constant, interparticle separation, coordination number, and particle size distribution data. The model is in reasonable agreement with experimental data when an adjustable interparticle separation parameter h_0 obtained from experimental data fitting is employed. Despite the agreement, there is no fundamental basis for the randomly variable separation for flocculated suspensions. Apart from interparticle forces, complicating suspension structures should also play a certain role in determining the network strength. Scales *et al.* (1997) advanced the model of Kapur *et al.* to describe the yield stress of suspensions as a function of charges in particle surface chemistry.

In this article we investigate the effects of volume fraction, particle size, and size distribution under constant surface chemistry conditions at the pH corresponding to the particle IEP. At the IEP, a maximum in viscosity and yield stress occurs and ionic strength does not affect the magnitude of the maximum yield stress although the pH range of flocculation is broadened at higher ionic strength [Leong *et al.*, (1991) and (1993)]. We focus here on the structural effects, separate from the surface chemistry, on the network strength of flocculated suspensions. This is achieved through scaling the experimental yield stress to the theoretical tensile strength. Links between them are explored and a simplified yield stress model is then provided. On the assumption that a proper statistic average diameter can represent the equivalent diameter of polydisperse particle systems for the characterization of the yield stress response, a number of models are proposed for size distributed suspensions and the suitability of the models is examined with a large quantity of experimental data. The aim is thus to characterize the effects of network structure in the system.

II. MONODISPERSE PARTICLE SYSTEMS

A. Basic concept

The phenomenon of yield stress is associated with a three dimensional, mechanically rigid particulate network structure that must deform and fail before flow can occur. The strength of the network structure determines the magnitude of yield stress τ_y. It is obvious that the density of network bonds and junctions and the strength of the bond determine the total strength of the network structure, and hence the yield stress τ_y which can be well described with the following model reduced from that of Kapur *et al.* (1997) and Scales *et al.* (1997) for a monodisperse particle system at the isoelectric point, where the particle interaction is only governed by the van der Waals attractive forces and suspensions exhibit the maximum yield stress

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$$T_{y\max} = \frac{A\Phi C_N(\Phi)}{24\pi h_0^2 d}. \tag{1}$$

In this formulation $T_{y\max}$ is the maximum tensile strength or theoretical maximum yield stress, Φ is the volume fraction of particles, d is the particle diameter, A is the Hamaker constant of the colloidal material, h_0 is the surface separation between two particles, and $C_N(\Phi)$ is the mean coordination number.

This equation is based on the assumption that the shear yield stress is simply equivalent to the summation of all pairwise interparticle forces per unit area between the primary particles in the suspension. Although interparticle interactions are dominant factors in determining the network strength, such a supposition is questionable according to the recently developed fractal type analysis [Potanin *et al.* (1995), Potanin and Russel (1996)]. In this treatment, the aggregate structure of weakly flocculated colloidal dispersions is imagined as a series of networked chains of varying elasticity and rigidity. Among them, the weak links will inelastically deform under lower strain than those associated with the rigid networks. An alternative analysis is to view the system as a series of weakly interconnected aggregate structures compromising strongly interacting primary particles. Clearly the weak interfloc links in such a system would yield under a lower stress than those within the aggregate. In this case, an experimental yield stress is lower than that predicted by Eq. (1).

However, as solid concentration increases, particles are more crowded together and weak links diminish gradually. Moreover, more crowded particles will produce geometric resistance to the deformation of networks. When such a suspension is sheared, besides forces needed to break down interparticle bonds, additional forces are required to push some particles away from each other. As a result, the experimental yield stress will be higher than the prediction of Eq. (1), exhibiting a positive deviation.

Outlined above are complicated structural factors whose effects on the network strength are still poorly understood. In order to coordinate the experimental yield stress and the theoretical tensile strength, a dimensionless yield stress, the ratio of measured shear yield stress to the predicted tensile strength, is suggested in this article. It allows the structural effects, separate from the surface chemistry influences, to be examined. As analyzed above the dimensionless yield stress should be a function of the volume fraction Φ, perhaps also a function of particle size d and surface separation h_0. As d and h_0 can be scaled with each other, we have:

$$\tau_{y\max}/T_{y\max} = f(\Phi, d/h_0). \tag{2}$$

Although the strictly theoretical analysis on this function is still not available at present, some properties of the function can be analyzed qualitatively. First, the functional value should have a magnitude of $O(1)$ in a quite large concentration region. This means that the network strength of flocculated suspension in the region is dominantly determined by interparticle forces. Second, there should be a critical volume fraction Φ_c. At this value, the dimensionless yield stress is equal to 1, implying a balance between the negative deviation arising from the weak links and the positive deviation from the geometric resistance. When $\Phi < \Phi_c$, the structural effect of weak links prevails over that of geometric resistance and vice versa.

In the first part of this article, we will examine structural effects on the yield stress and determine the empirical function using experimental data. Some feasible explanations for these effects are then given.

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TABLE I. Surface area and particle size distributions of Al_2O_3 particles. (n = number, s = surface area, v = volume average particle size; d_{10}, d_{50} and d_{90} are size cutoffs of the particles by surface area lie below 10%, 50%, and 90%, respectively).

Samples	BET (m^2/g)	d_n (μm)	d_s (μm)	d_v (μm)	d_{90} (μm)	d_{50} (μm)	d_{10} (μm)	$\frac{d_v}{d_n}$
AKP-15	3.8	0.369	0.580	0.690	0.945	0.548	0.267	1.87
AKP-20	4.3	0.208	0.401	0.558	0.761	0.332	0.147	2.68
AKP-30	6.8	0.146	0.247	0.326	0.451	0.169	0.109	2.23
AKP-50	10.5	0.135	0.185	0.250	0.391	0.140	0.107	1.85

B. Experimental materials and methods

Four Al_2O_3 particulate samples were used in this study. They were supplied by Sumitomo Chemicals Pty. Ltd of Japan. The particles have a density of 4.0 g/cm^3, an isoelectric point of about pH 9, and are nearly spherical. The particle size distributions measured by a laser diffraction technique with a Coulter Counter LS 130 performed in this study and the Brunauer–Emmett–Teller (BET) surface area data reported by Sumitomo Chemicals are shown in Table I. The number, volume, and surface area average particle sizes were determined by the standard algorithm in the Coulter software and will be explained in more detail in Sec. III of this article.

The electroacoustic technique was also applied for particle sizing using a Matec Acousto-Sizer as a reference to the results obtained through laser diffraction. The differences between d_s measured with the Coulter and the mean diameters measured with the Matec Acousto-Sizer were less than 10%. The ratio data of d_v to d_n of the four particles indicate the distributions of particle size are rather narrow although they can not be strictly regarded as monodisperse. Various broad size distributed suspensions were obtained by blends of particles of two or more different sizes in proper proportions.

Suspensions of various solid concentrations were prepared by adding distilled water to the particle samples. They were then sonicated for about 2 min with a high intensity sonic probe (Sonifier B30, 20 kHz, 350 W) to break up any large lumps or granules in the suspension and to create a smooth flocculated suspension. The suspensions were then allowed to rest for a few days to establish chemical and physical equilibrium prior to any measurements. Concentrated nitric acid and potassium hydroxide (1–5 M) were used to adjust suspension pH. The salt concentration is about 0.01 M. The yield stress was measured at temperature 20 ± 1 °C with the Haake RV3 rheometer using the vane technique developed by Nguyen and Boger (1985). A Radiometer PHM82 standard pH meter was employed to measure suspension pH.

C. Results and discussion

The effects of the volume fraction and particle size on the maximum yield stress for the suspensions of four narrow size distributed particles are shown in Fig. 1. The $\tau_{y\max}$ data were measured at pH8.9 ± 0.2, the vicinity of IEP, with good reproducibility. It can be seen in Fig. 1 that the $\tau_{y\max}$ is extremely sensitive to both volume fraction Φ and particle size d, increasing very sharply as Φ increases and d decreases.

Using experimental data in combination with the prediction of Eq. (1), the dimensionless yield stress was determined for the four alumina particles. The results are shown in Fig. 2. The parameters used here were surface separation $h_0 = 24$ Å as determined by Johnson (1998) assuming independence of particle sizes, and a Hamaker constant of

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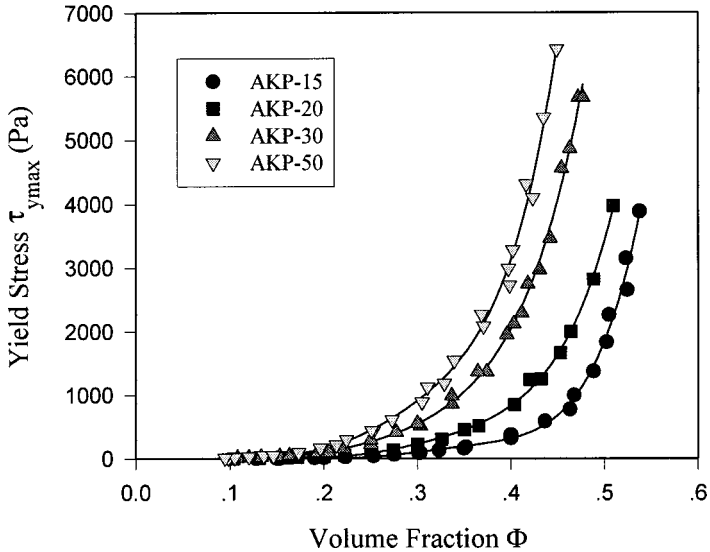

FIG. 1. The maximum yield stress of alumina suspensions versus volume fraction. The results were measured at pH 8.9±0.2.

5.3×10^{-20} [Hough and White (1980)]. A number of theoretical and empirical equations have been proposed for relating coordination number with volume fraction of monodisperse spheres [Iwata and Homma (1974); Suzuki *et al.* (1981)]. First, the relationship of Gotoh cited by Suzuki *et al.* (1981) and used by Kapur *et al.* (1997):

$$C_N(\Phi) = 36\Phi/\pi, \quad \Phi \leqslant 0.47 \tag{3}$$

was employed. To clearly show the effects of volume fraction and particle size, only the data under 0.43 volume fraction are included. It can be seen that the value of dimensionless yield stress is between about 0.1 and 5, indicating a magnitude of order one. This

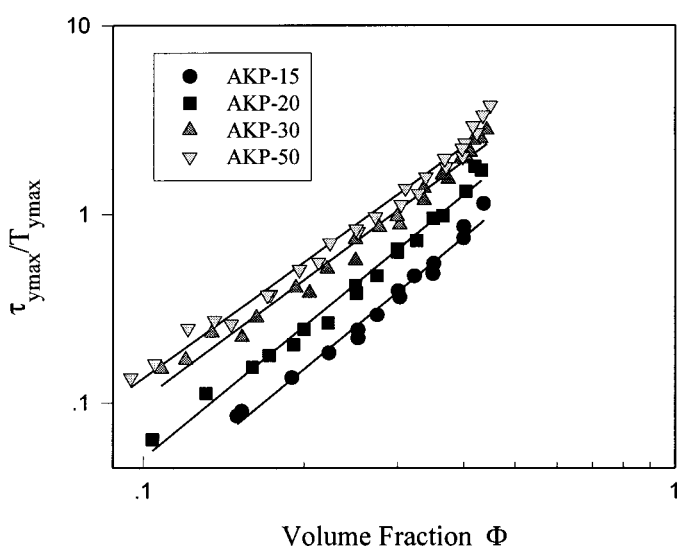

FIG. 2. The dimensionless yield stress as a function of volume fraction.

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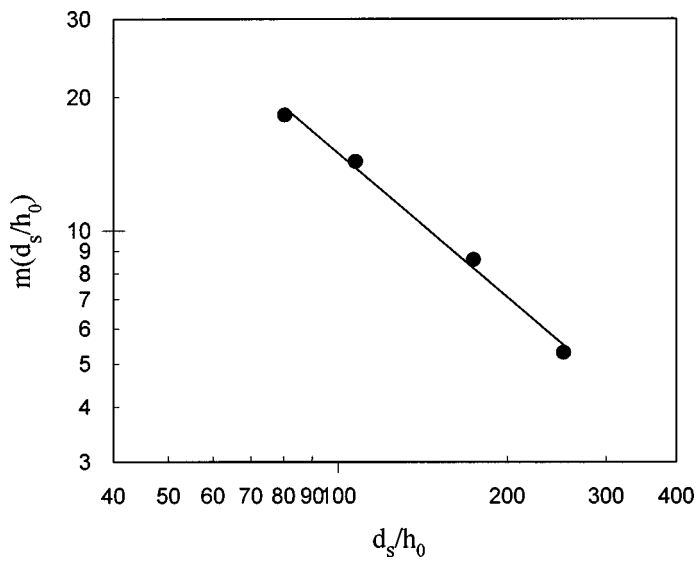

FIG. 3. The fitting constant $m(d_s/h_0)$ in Eq. (4) as a function of d_s/h_0.

verifies that the tensile strength model gives a proper prediction of magnitude order of yield stress and the network strength is dominantly determined by interparticle interactions in the concentration region ($\Phi < 0.43$).

From Fig. 2, we can see (on a log–log scale), that the four curves are roughly linear and nearly parallel, indicating the influences of concentration and particle size on the dimensionless yield stress are independent. Assuming a power-low relation between the dimensionless yield stress and volume fraction, the average exponent obtained is about 2.2 through regression technique. Using the relation

$$\tau_{y\max}/T_{y\max} = f(\Phi, d/h_0) = m\Phi^{2.2}, \tag{4}$$

where $m = f(d/h_0)$, to fit all data, and plotting the fitting results of $m(d/h_0)$ against d/h_0 on a log–log scale, a straight line with a slope of minus 1 is obtained as shown in the Fig. 3, suggesting the dimensionless yield stress is inversely proportional to particle size. This is somewhat surprising, as the dimensionless yield stress should be independent of the particle size if the model (theoretical tensile strength) accounts for all effects due to particle size. The model [Kapur *et al.* (1997)] incorporates an inverse square size dependence for the number of particle contacts per unit area, and a direct dependence on size for the force between adjacent particles. This results in the inverse dependence on particle size as indicated in Eq. (1). The inverse size dependence of the dimensionless yield stress observed suggests that the experimentally measured yield stress has an additional inverse size dependence not accounted for in the model. Although there is currently no explanation for this behavior it is not likely to come from the force dependence (which is well predicted by traditional DLVO equations). It is possible that either the structural effects discussed in this article or nonsphericity of the particles may be responsible for the extra inverse size dependence observed.

There is still no convincing explanation for the structural effect of the particle size dependency on the yield stress. It could be argued that the spatial structure of suspensions for the monodisperse particle systems of different size should be the same, provided that h_0 varies with particle size and d/h_0 remains constant. This might be true for well

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dispersed suspensions. However, the spatial structure might be different if h_0 is constant for the flocculated suspensions of different particle sizes. In this case, the effective volume occupied by flocs or aggregates in the suspension could be higher for smaller particles than for larger ones at a given solid concentration. Thus, the dimensionless yield stress would increase as d/h_0 decreases.

Comparing this result with the theoretical prediction, a simplified equation is obtained in this study whereby the additional inverse size dependence is included in the yield stress model, such that:

$$\tau_{y\max} = K\Phi^c/d^2, \tag{5}$$

where

$$K = Ab/24\pi h_0, \tag{6}$$

and b and c are fitting parameters depending on the solid concentration regime.

The products of maximum yield stress multiplying the square of particle size against volume fraction are plotted in Figs. 4(a)–4(c) using different statistic average sizes. In all three figures, the original four curves in Fig. 1 are reduced to a single curve, demonstrating an inverse square dependence of $\tau_{y\max}$ on particle size, as suggested by Eq. (5). Other researchers also observed a similar dependence of shear or compressive yield stress on particle size using some percentile size, such as d_{50} or d_{10} [Buscall *et al.* (1988); Leong *et al.* (1995); Liddell (1996); Miller *et al.* (1996)]. However, some data were rather scattered because the particle size data (d_{50} or d_{10}) did not include relevant information of the size distribution. For close to monodisperse particles, we consider that a statistical average size should characterize the effects of both particle size and size distribution on the yield stress. The results in Figs. 4(a)–4(c) are insufficient to determine which average size is the most suitable, which is attributed to the fact that the four alumina particles have a rather narrow particle size distribution and the difference among three average sizes is thus small. However, the better fit likely suggests that the surface area average diameter is the best. In Sec. III, we will further examine size distribution effects in detail and confirm that the surface area average diameter d_s is the most suitable.

Figure 5 is a log–log plot of $\tau_{y\max}\, d_s^2$ vs Φ. A linear relationship with varied slopes in different concentration regions is obtained, indicating a power-law relationship between $\tau_{y\max}$ and Φ first suggested by Thomas (1961). Using a regression technique, the curve has a best fit slope of 4.2 in the range $\Phi < 0.42$ and a slope of 8.5 at $\Phi > 0.42$. A transition point appears at a volume fraction of about 0.41–0.43. Above the transition point, the maximum yield stress exhibits a much more pronounced concentration dependence. Many groups have recently observed a similar dependence of yield stress on Φ such as an exponent of 4 for zirconia suspensions [Leong *et al.* (1995)], 3.8 for iron oxide suspensions [Lin *et al.* (1991)], 4.5 for TiO_2 pigment suspensions [Liddell (1996)] and 5 for alumina suspension of AKP-15 [Channell and Zukoski (1997)]. Thomas (1961) obtained an exponent of 3 for suspensions of thorium oxide, graphite, and kaolin. Buscall (1987) and Alderman *et al.* (1991) also obtained an exponent of 3 for suspensions of polystyrene lattice and bentonite clay, respectively. de Krester (1995) found that the exponent varied between 4.59 and 9.44 for suspensions of coal mine tailings.

It is important to note that, in some of the above studies, surface chemistry conditions were not well defined, surface charge was heterogenous in some instances, and the effect of size distributions was not considered. However, these suggestions alone may not be enough to account for the observed deviation between results. It is expected that although

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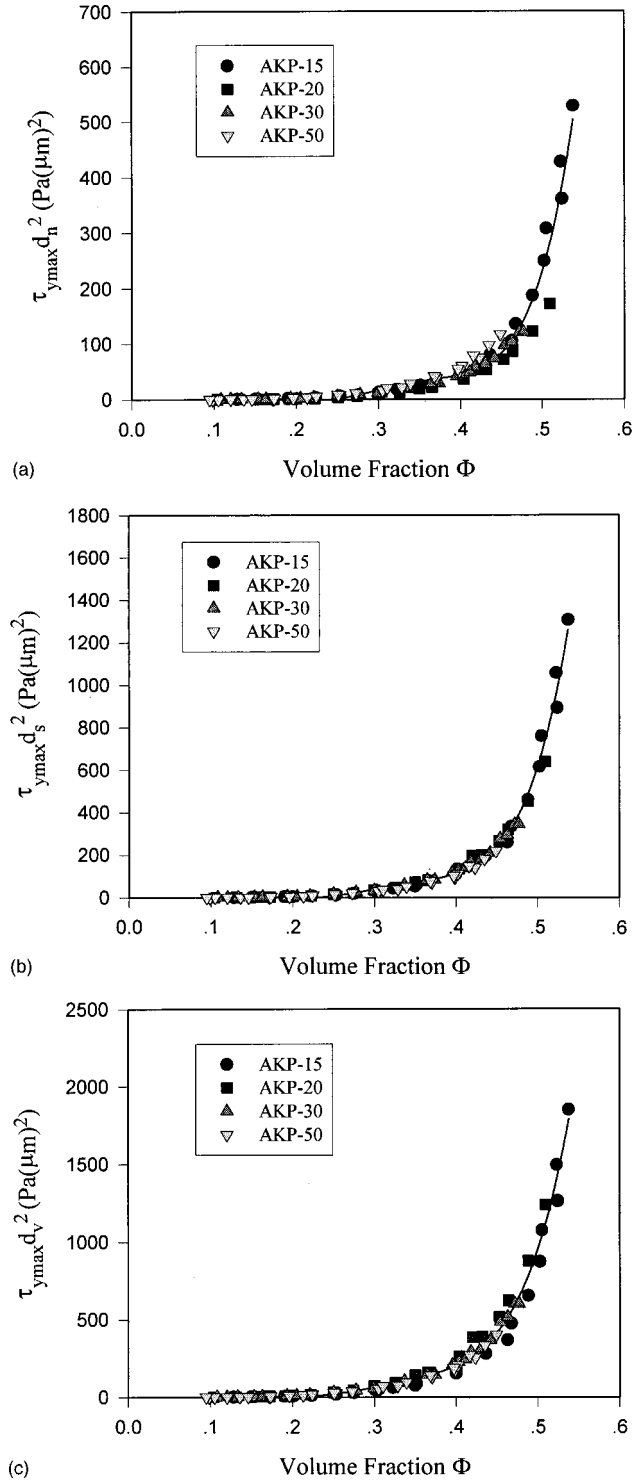

FIG. 4. The products of maximum yield stress multiplying d_n^2(a), d_s^2(b) and d_v^2(c) as a function of volume fraction.

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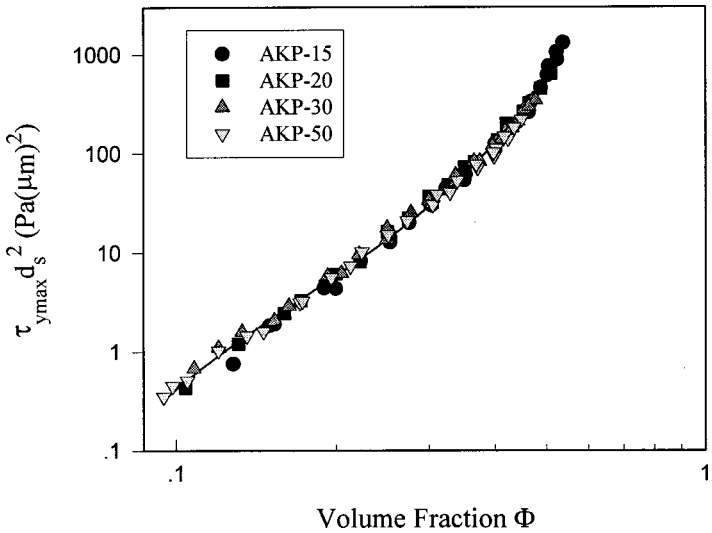

FIG. 5. A log–log plot of $\tau_{y\max} d_s^2$ against volume fraction.

a suspension will show a variety of network linkages, the population of strong linked bonds and junctions will play the most important role in determining the network strength. The strength will then increase with the elimination of weakly linked bonds and junctions as the particle crowding or the volume fraction of solids Φ increases. When a certain threshold of Φ is reached, the weak and low coordination linkages will be eliminated, and meanwhile the geometric resistance will become more pronounced. This is another way of saying that the disorder flocculated structure will become more regular and orderly as the solid concentration increases. It will reach a crystal-like structure at high volume fraction. Eventually, when a suspension is close to the maximum packing volume fraction, every particle lies in contact with all the nearest neighbor particles. Therefore, the contribution of structural effects to the network strength can be compared to that of interparticle interactions and the yield stress should exhibit a much higher dependence on Φ as the solid loading overtakes a certain threshold. We postulate that the transition point in the log–log plot of $\tau_{y\max} d_s^2$ vs Φ observed in this study corresponds to this threshold.

On the other hand, there should exist a low concentration threshold, under which the suspension is in a structure of weak and low coordination linkages with little or no geometric resistance. The significant negative deviation between the measured yield stress and the prediction may arise, which generates a dimensionless yield stress far smaller than unity. Under these conditions the yield stress should exhibit a weaker dependence on Φ with a lower exponent c. It is difficult to verify this prediction experimentally by means of the measurements using the vane technique due to experimental limitations, such as settling at low concentrations and instrumental errors at low yield stress regions.

The power-law relationship between τ_y and Φ can be in connection with the fractal model of Potanin and co-workers [Potanin *et al.* (1995); Potanin and Russel (1996)] through the relationship between the exponent c and the fractal dimension d_f. The relationship is described as $c = 3/(3\text{-}d_f)$. The average exponent of 4.2 observed gives a fractal dimension of 2.29 in the volume fraction range smaller than around 0.43. Above

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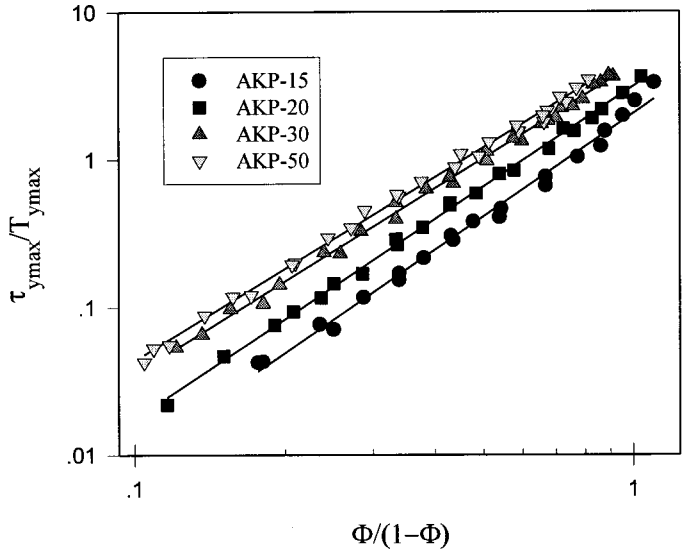

FIG. 6. The dimensionless yield stress as a function of $\Phi/(1-\Phi)$.

it, the average exponent of 8.5 gives a fractal dimension of 2.65. The change in fractal dimension suggests a change of the network structure of the suspension as detailed above.

Logically, the functional forms of both the theoretical tensile strength and the dimensionless yield stress will be changed depending on what coordination number function $C_N(\Phi)$ is chosen. If the relationship of Rumpf cited by Suzuki *et al.* (1981):

$$C_N(\Phi) = 3.1/(1-\Phi) \tag{7}$$

is employed, the dependence of tensile strength on Φ takes a functional form of $\Phi/(1-\Phi)$ as suggested by Rumpf (1962) and Eq. (1) becomes:

$$T_{y\max} = \frac{3.1A\Phi}{24\pi h_0^2 d(1-\Phi)}. \tag{8}$$

We assume that the dependence of dimensionless yield stress on Φ takes a similar functional form and the results of dimensionless yield stress against $\Phi/(1-\Phi)$ on a log–log plot are shown in the Fig. 6. Four parallel linear curves are obtained. Applying regression analysis combined with the prediction of Eq. (8), an alternative dimensionless yield stress function with fitting constant $B = 490$ is obtained such that

$$f(\Phi, d/h_0) = B\left(\frac{\Phi}{1-\Phi}\right)^{2.2}\frac{h_0}{d}. \tag{9}$$

Thus, an alternative simplified yield stress function can be given as

$$\tau_{y\max} = K\left(\frac{\Phi}{1-\Phi}\right)^{c}\frac{1}{d^2}, \tag{10}$$

where

$$K = \frac{3.1Ab}{24\pi h_0}, \tag{11}$$

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TABLE II. Φ_c values for the suspensions of different particle size.

	AKP-15	AKP-20	AKP-30	AKP-50
Φ_c	0.445[a]/0.425[b]	0.376/0.385	0.302/0.344	0.265/0.305

[a]Determined by Eq. (4).
[b]Determined by Eq. (9).

and the fitting parameters b and exponent c remain constant in the solid volume fraction varying from 0.1 to 0.53. In this case, the values of b and c are 490 and 3.2, respectively.

From Eqs. (4) and (9), the critical volume fraction Φ_c was calculated by setting $f(\Phi, d/h_0) = 1$. The results are shown in Table II. Although the values of Φ_c are slightly different using different equations, the tendency is clear. The Φ_c decreases with decreasing particle size as explained previously. This suggests that the volume fraction required for the balance of negative deviation arising from weak linkages in the network structure and the positive deviation from geometric resistance decreases as particle size decreases.

Overall, a strictly theoretical analysis on the structural contributions to the network strength of flocculated suspensions is not available at present, and the empirical functional form of the dimensionless yield stress depends on what functional form is chosen for the coordination number. Structural effects may also arise from the nonspherical shape of particles. Nonspherical particles will generate a more significant geometric resistance to the deformation of suspensions under shear or compression than spheres. In addition, the functional form of coordination number should be changed for nonspherical particles. More work is needed to understand this effect clearly. The study nonetheless verifies that the theoretical tensile strength model [Kapur *et al.* (1997); Scale *et al.* (1997)] provides a proper prediction of the magnitude of yield stress and that interparticle forces play a dominant role in determining the network strength over a quite wide solid concentration region. It is suggested that deviations between the experimental yield stress and theoretical prediction are attributed to the structural effects controlled by the competition between weak linkages and geometric resistance on the network strength. Above the transition point, around $\Phi = 0.41-0.43$, the effect of geometric resistance to the deformation on the network strength of suspension is postulated to be more pronounced. In summary, therefore, combining the effects of interparticle interactions and structural effects, the experimental yield stress depends on the inverse of the square of particle size and can be related to $\Phi/(1-\Phi)$ using a power law relation or related to Φ with a varied exponent in different concentration regime.

III. POLYDISPERSE SYSTEMS

A. Basic concept

There are a number of ways of dealing with polydisperse systems that are typical of many suspensions. A simple assumption is that there exists an equivalent diameter where the suspension has exactly the same yield stress as the monodisperse suspension of the same size. This diameter will certainly be dependent on the particle size distribution and mostly could be an appropriate statistic average diameter. Three commonly used statistic average diameters are the number d_n, surface area d_s, and volume average diameter d_v. They have the same value for monodisperse systems but follow the relationship of $d_n < d_s < d_v$ for size distributed particles. The difference among their values characterizes the polydispersity of particles. These three diameters are defined as follows:

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$$d_n = \sum \left(n_i d_i \Big/ \sum n_i \right), \tag{12}$$

$$d_s = \sum \left(s_i d_i \Big/ \sum s_i \right), \tag{13}$$

$$d_v = \sum \left(v_i d_i \Big/ \sum v_i \right), \tag{14}$$

where n_i, s_i, and v_i, are, respectively, the number, surface area, and volume of the ith particles which have the diameter of d_i. The above equations can be interchanged such that:

$$d_n = \sum \frac{x_{vi}}{d_i^2 \Sigma (x_{vi}/d_i^3)}, \tag{15}$$

$$d_s = 1 \Big/ \sum (x_{vi}/d_i), \tag{16}$$

$$d_v = \sum x_{vi} d_i, \tag{17}$$

where $x_{vi} = v_i / \Sigma v_i$ is the volume fraction ratio of particles of d_i. Either Eq. (5) or Eq. (10) can then be rewritten as:

$$\tau_{y\max} = F(\Phi)/d^2. \tag{18}$$

Replacing d in the above equation with Eqs. (15)–(17) gives the following three equations for the prediction of the yield stress of polydisperse systems at the IEP:

$$\tau_{y\max} = \frac{F(\Phi)}{\left(\Sigma \dfrac{x_{vi}}{d_i^2 \Sigma (x_{vi}/d_i^3)} \right)^2}, \tag{19}$$

$$\tau_{y\max} = F(\Phi) \left(\sum \frac{x_{vi}}{d_i} \right)^2, \tag{20}$$

$$\tau_{y\max} = \frac{F(\Phi)}{(\Sigma x_{vi} d_i)^2}. \tag{21}$$

Considering $\tau_{yi} = F(\Phi)/d_i^2$, the above three equations can be changed to:

$$\tau_{y\mathrm{mix}} = \left(\sum \frac{x_{vi}}{\tau_{yi}^{1/2} d_i^3 \Sigma (x_{vi}/d_i^3)} \right)^{-2}, \tag{22}$$

$$\tau_{y\mathrm{mix}} = \left(\sum x_{vi} \tau_{yi}^{1/2} \right)^2, \tag{23}$$

$$\tau_{y\mathrm{mix}} = \left(\sum x_{vi} / \tau_{yi}^{1/2} \right)^{-2}, \tag{24}$$

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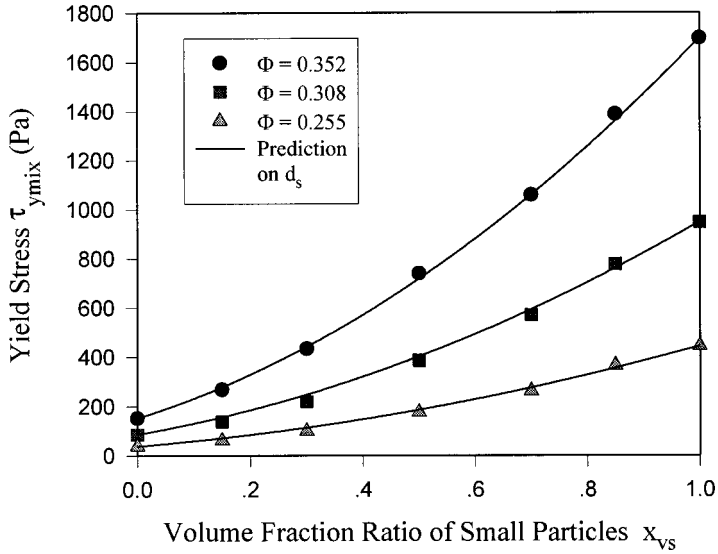

FIG. 7. The comparison of measured yield stress with the prediction of Eq. (23) for bimodal suspensions (AKP-15+AKP-50) at different concentrations.

where $\tau_{y\text{mix}}$ is the yield stress of the mixture suspension and τ_{yi} and x_{vi} are the yield stress and volume fraction ratio of each individual suspension. Equations (19) and (22) are deduced on the basis of number average diameter, Eqs. (20) and (23) on the surface area, and Eqs. (21) and (24) on the volume average diameter. It is not expected that all of the above equations will predict the actual behavior of polydisperse suspensions. In this context, we now examine which average diameter is the most suitable.

B. Results and discussion

The yield stress of many suspensions of broad distributed sizes with blends of two, three, or four kinds of particles in varied proportions was measured and predicted using Eqs. (22), (23), and (24) to examine which is the most suitable. Figures 7 and 8 show the measured and predicted yield stress using Eq. (23) on the surface area average diameter d_s versus the volume fraction ratio of small particles for the bimodal suspensions of AKP-15+AKP-50, and AKP-15+AKP-30 at various concentrations. Very good agreement between experimental data and prediction is obtained. As a comparison, the measured and predicted yield stress with Eqs. (22) and (24) are shown respectively in Figs. 9 and 10, versus the volume fraction ratio of small particles. It is observed that the predicted yield stress with Eq. (22) on d_n is higher than the measured value. Because the d_n overweights the contribution of small particles to the statistic average diameter it gives the smallest average particle size value, compared to the other two statistic average diameters. It will also overweight the contribution of small particles to the predicted yield stress when the smallest average size value is used in the relationship of reverse square particle size, thus showing us a higher predicted value. Contrary to d_n, the predicted yield stress with Eq. (24) based on the diameter d_v which overweights the contribution of large particles and gives the largest average particle size value is lower than the true value. This result is shown in Fig. 10. However, d_s provides a balance between the contributions of small and large particles to the average size.

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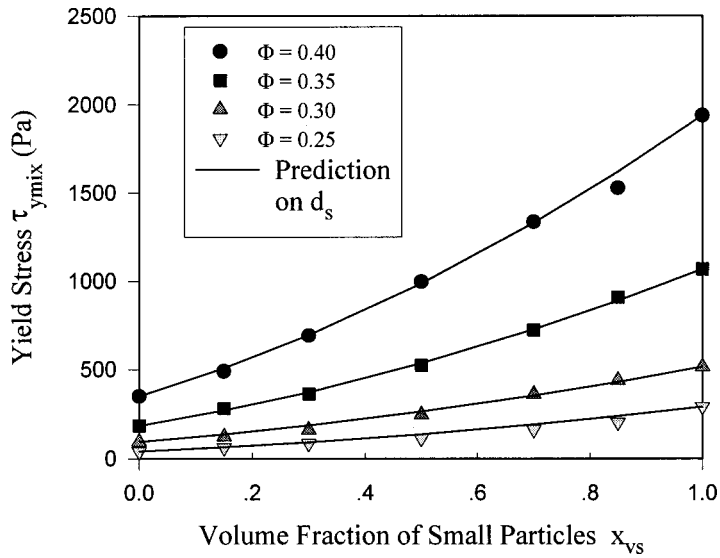

FIG. 8. The comparison of measured yield stress with the prediction of Eq. (23) for bimodal suspensions (AKP-15+AKP-30).

The results of measured and predicted yield stress for the blend suspensions of trimodal systems are given in Table III. Figure 11 compares the measured yield stress data of all blend suspensions including bi-, tri-, and tetramodal systems with predicted values using Eq. (23). The blend suspensions are of quite different size distribution and polydispersity, including close to single sized, left skewed, right skewed, or nearly normalized distributed particles. The good agreement for such a range of suspensions demonstrates that the surface area average diameter d_s can be used to quantitatively characterize the

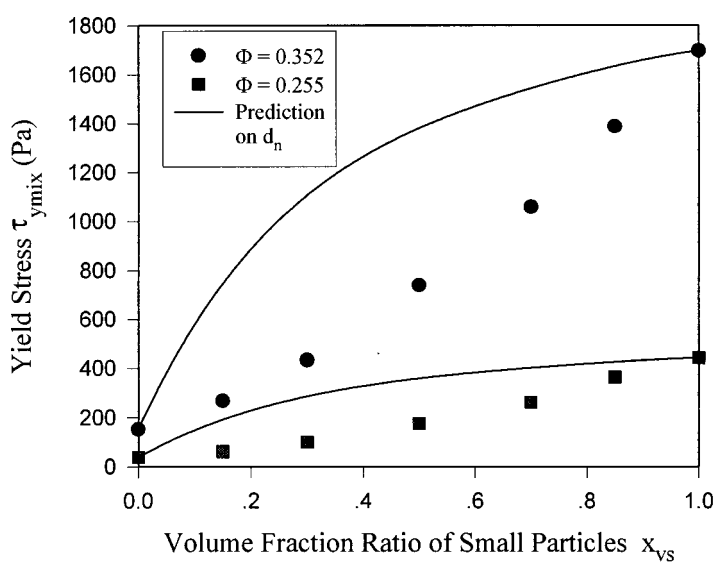

FIG. 9. The comparison of measured yield stress with the prediction of Eq. (22) for bimodal suspensions (AKP-15+AKP-50).

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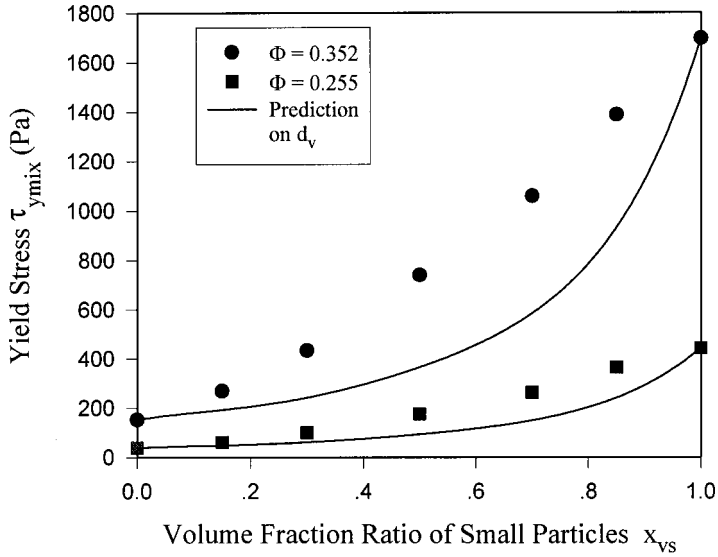

FIG. 10. The comparison of measured yield stress with the prediction of Eq. (24) for bimodal suspensions (AKP-15+AKP-50).

effect of particle size distributions on the yield stress of flocculated suspensions. Figure 12 shows the comparison of measured yield stress with predictions using Eqs. (22) and (24) for all blend suspensions. The poorer agreement indicates d_n and d_v are not suitable parameters to relate the relationship between yield stress and size distribution.

The relationship between the yield stress and different statistic average diameters for blend suspensions of AKP-15 and AKP-50 is shown in Fig. 13, as the log–log plot of $\tau_{y\mathrm{mix}}/\Phi^{4.2}$ vs d_n, d_s, and d_v. The diameters of d_n, d_s, and d_v were calculated using Eqs. (15), (16), and (17) with the size distribution data of AKP-15 and AKP-50. Figure 13 clearly shows that a linear relationship only exists between $\tau_{y\mathrm{mix}}/\Phi^{4.2}$ and d_s (not for d_n and d_v) and the slope of the plot is about -2. Similar results were obtained for blends of AKP-15 and AKP-30. It is important to point out (as shown in Fig. 13) that a different dependency of yield stress on particle size rather than the exponent value of -2 could be obtained if other particle size parameters such as d_n or d_{50} are employed for those

TABLE III. Values of measured and predicted yield stress for trimodal blends [total volume fraction $\Phi_T = 0.306$; $\tau_1 = 87$ (Pa), $\tau_2 = 196$ (Pa), $\tau_3 = 962$ (Pa)].

AKP-15 x_{v1}	AKP-20 x_{v2}	AKP-50 x_{v3}	Measured yield stress (Pa)	Predicted yield stress (Pa) on d_s	on d_v	on d_n
0.10	0.10	0.80	732	737	525	867
0.15	0.20	0.65	577	594	380	771
0.25	0.25	0.50	449	456	271	677
0.30	0.35	0.35	351	344	213	546
0.25	0.55	0.20	268	264	190	384
0.21	0.62	0.17	257	253	191	350
0.40	0.45	0.15	212	216	157	346
0.55	0.35	0.10	167	172	132	296
0.75	0.20	0.05	133	129	108	224

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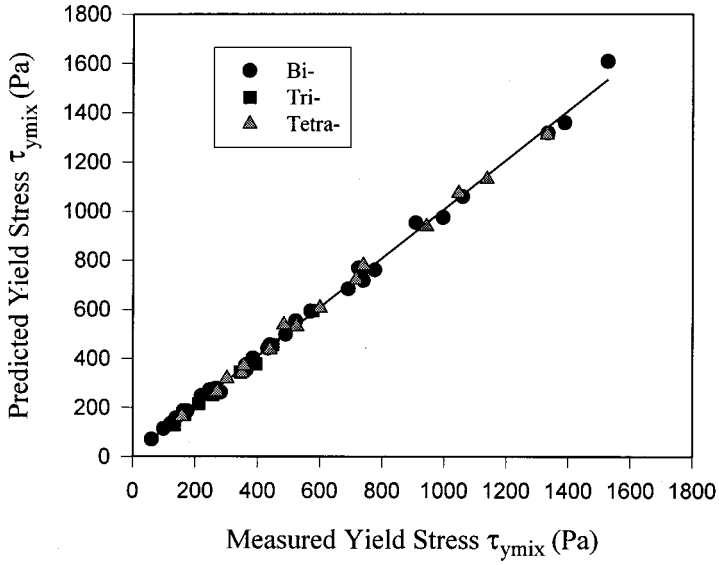

FIG. 11. The comparison of measured yield stress with the prediction of Eq. (23) for bi-, tri-, and tetramodal suspension systems.

suspension systems. Figure 14 shows a log–log plot of $\tau_y/\Phi^{4.2}$ vs d_s. The data include that obtained with narrow size distributed suspensions and broad size distributed blend suspensions. In this graph, a power-law relationship is clearly evident. Using a regression technique, the linear curve has a best fit slope of -2.02.

It is well established that, for a dispersed hard sphere suspension, using broad size

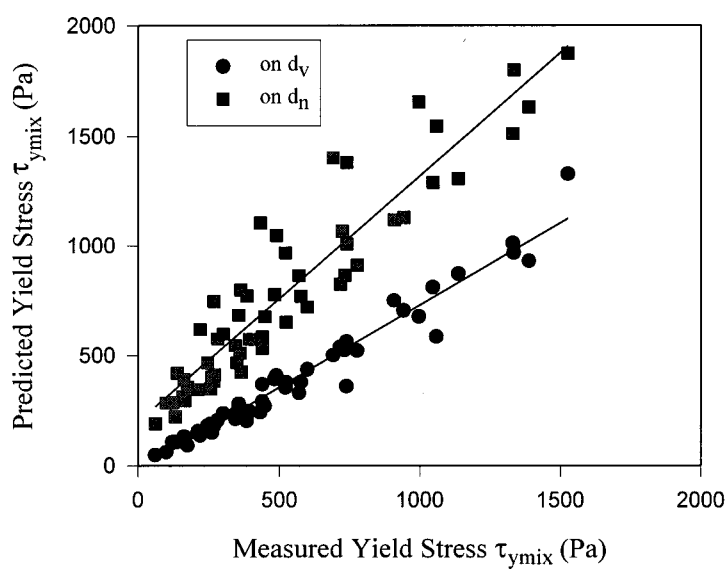

FIG. 12. The comparison of measured yield stress with the prediction of Eq. (22) (■) and Eq. (24) (●) for bi-, tri-, and tetramodal suspension systems.

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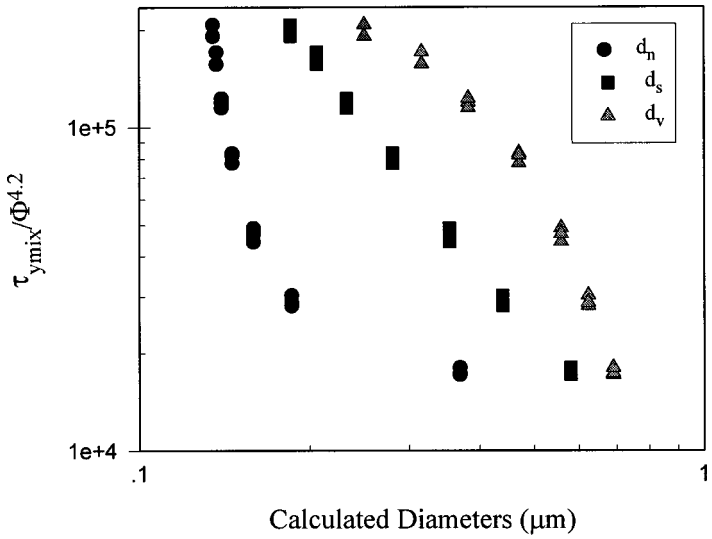

FIG. 13. The yield stress vs different statistic average diameters for bimodal suspensions (AKP-15+AKP-50) at various solid concentration.

distributed particle systems allows the viscosity to be reduced while maintaining the same solid loading. However, for a flocculated concentrated suspension, the conclusion is exactly the opposite.

As an example of the mixing model postulated here, three monodisperse particles with diameters of d_1, d, and d_2 such that $d_1 < d < d_2$ and the size ratio $r_1 = d_1/d$ and $r_2 = d_2/d$, a bimodal suspension which has the same volume average size as d can be prepared by mixing suspensions of d_1 and d_2 if the following equations are met:

$$d = x_{v1}d_1 + x_{v2}d_2, \tag{25}$$

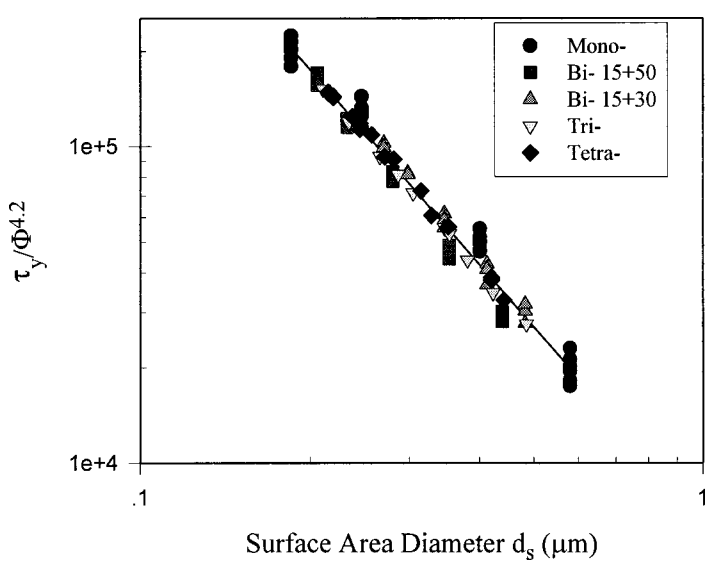

FIG. 14. The relationship between yield stress and particle size.

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and

$$1 = x_{v1} + x_{v2}. \tag{26}$$

Solving the above equations yields:

$$x_{v1} = \frac{r_2 - 1}{r_2 - r_1} \tag{27}$$

and

$$x_{v2} = \frac{1 - r_1}{r_2 - r_1}. \tag{28}$$

The yield stress of the bimodal suspension $\tau_{y\text{-bi}}$ calculated with Eq. (20) is:

$$\tau_{y\text{-bi}} = \frac{F(\Phi)}{d^2}\left(\frac{r_1 + r_2 - 1}{r_1 r_2}\right)^2. \tag{29}$$

Noticing that $F(\Phi)/d^2$ is equal to the yield stress of a monodisperse suspension of particle size d, here signed as $\tau_{y\text{-mono}}$, the yield stress ratio R_τ of the bimodal suspension to the monodisperse suspension is

$$R_\tau = \frac{\tau_{y\text{-bi}}}{\tau_{y\text{-mono}}} = \left(\frac{r_1 + r_2 - 1}{r_1 r_2}\right)^2. \tag{30}$$

Because of $r_1 < 1$ and $r_2 > 1$ it is easy to demonstrate that R_τ is larger than unity. The R_τ will increase with an increase in the polydispersity of the bimodal suspension either through decreasing r_1 or increasing r_2. For example, the yield stress of a bimodal suspension should be 2.25 times higher than that of a monodisperse suspension of the same d_v for the system of $r_1 = 1/2$ and $r_2 = 2$, but 5.44 times higher for the system of $r_1 = 1/3$ and $r_2 = 3$. This suggests that the broad size distributed suspension will have a higher yield stress than the narrow size distributed suspension of the same volume average diameter. At the same volume average diameter, the broad size distributed particles always have the lower values of d_s and d_n, and higher particulate number concentration.

A specific experiment was designed to confirm this concept. A mixture of AKP-50 ($x_{v1} = 0.827$) and AKP-15 ($x_{v2} = 0.173$) produced a blend suspension which has the same d_v as AKP-30 but with a broader size distribution. Figure 15 compares the yield stress-pH behavior of the blend suspension with AKP-30 at $\Phi = 0.3$. The narrow size distributed AKP-30 has a $\tau_{y\max}$ of 530 Pa, while the broader size distributed blend suspension has a $\tau_{y\max}$ of 690 Pa. The yield stress ratio R_τ obtained from experimental data is 1.3 which is very close to the value of 1.35 calculated by Eq. (30) for the experimental systems, where the r_1 is 0.767 and r_2 is 2.117.

On the other hand, a blend suspension of AKP-50 and AKP-15, which has the same d_s as AKP-30, was prepared to compare its yield stress–pH behavior with that of AKP-30. The results shown in Fig. 16 illustrate only a small difference in the $\tau_{y\max}$ value between the two suspensions. The agreement between these two samples again demonstrates that d_s can be used to characterize the effect of particle size and size distribution on the rheological behavior of flocculated concentrated suspensions.

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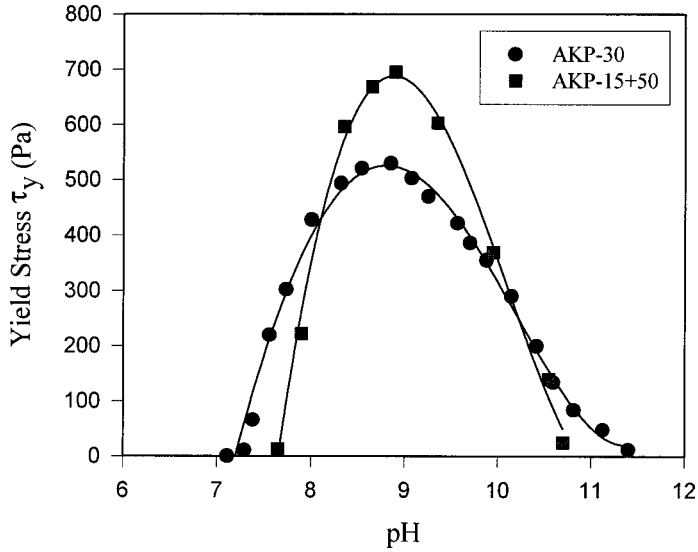

FIG. 15. The yield stress–pH behavior of two alumina suspensions (same d_v) subjected to different size distributions.

IV. CONCLUSION

In this article we have explored the structural effects on the network strength of suspensions, separate from the surface chemistry influences, by comparing experimental data with theoretical predictions from the tensile strength model. We find that the tensile strength model provides proper prediction of the magnitude order of the yield stress below a certain volume fraction (around 0.41–0.43 in this case) and that the complicating structural effects arising from the competition of the influences between the weak linkages and geometric resistance on the network strength cause a deviation between predic-

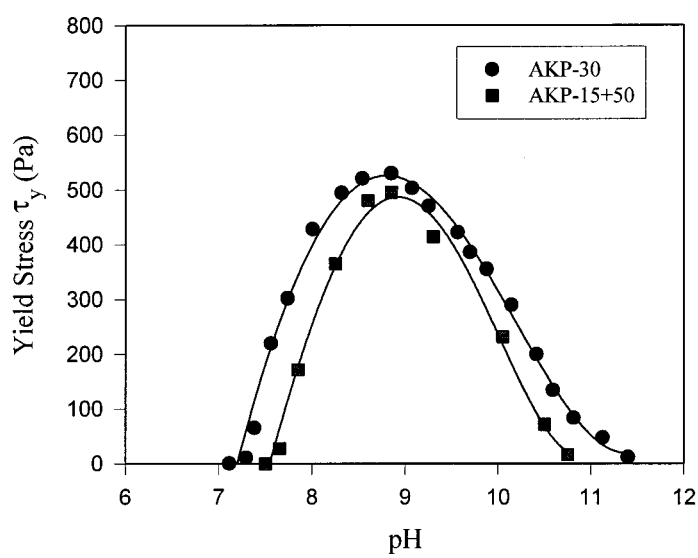

FIG. 16. The yield stress–pH behavior of two alumina suspensions (same d_s).

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tion and measurement. At higher volume fraction, the effect of geometric resistance on the deformation of suspensions becomes more pronounced. The experimental yield stress is found to depend on the inverse square of the particle size and can be related to $\Phi/(1-\Phi)$ using a power-law relation or with Φ with a varied exponent. Having examined a number of proposed models for polydisperse suspensions with experimental data, we demonstrate that the surface average diameter d_s can characterize the effects of particle size and size distributions on yield stress response of concentrated flocculated suspensions. A broad size distributed suspension exhibits a higher yield stress than a narrow size distributed suspension of the same volume average diameter at a given solid concentration.

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